1. Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy (Point 13 in the list)

This is the core ability for solving this problem. Students need to understand the real-world constraints, specifically that it's impossible to precisely calculate the area of an irregular shape (like an oil spill) using standard geometric formulas (such as for rectangles or triangles). Therefore, they need to use an **estimation** method.

- **Specific Analysis:** The question asks to "estimate" the area of the oil spill. Without grid paper, the most common estimation methods involve decomposing the irregular shape into more regular approximate shapes (e.g., rectangles and triangles) or using an overlaid grid counting method. Students need to use visual judgment and approximate measurements based on the map scale (1 cm represents 10 km), realizing that the result will be an estimate, not a precise value. This understanding of the appropriate level of accuracy in a real-world context guides the choice of solution strategy.

2. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Point 7 in the list)

This task requires students to manipulate multiple types of information.

- **Specific Analysis:** Students need to extract information from the graphical representation (the map), perform numerical manipulation using the scale (converting centimeters to kilometers, then calculating square kilometers), and represent the result as a final area value. This involves switching between different representations and manipulating data.

3. Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation (Point 12 in the list)

The estimation process involves multiple steps.

- **Specific Analysis:**
  1. Choose an estimation method (e.g., decomposing the shape into a few approximate regular shapes).
  2. Visually measure or estimate the dimensions of these approximate shapes on the map in centimeters.
  3. Apply the scale to convert centimeter dimensions to kilometer dimensions.
  4. Use area formulas to calculate the actual area of each approximate shape in square kilometers.
  5. Add all the approximate areas to get the total estimated area.
     This is a clear multi-step procedure.

4. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Point 5 in the list)

Students need to understand the basic mathematical definitions and rules for area and scale.

- **Specific Analysis:** Students must understand how the scale affects the conversion of area units ($1\ \mathrm{cm}^2$ represents $10\ \mathrm{km} \times 10\ \mathrm{km} = 100\ \mathrm{km}^2$). They need to know the formulas for calculating the area of a rectangle or triangle and apply these basic algorithms during the estimation process.

1. 利用对背景的理解来指导或加快数学解决过程，例如，在背景适当的准确度水平上进行工作 (列表中的第 13 点)

这是解决此问题的核心能力。学生需要理解现实世界的限制，即无法通过标准的几何公式（如矩形或三角形面积）精确计算不规则形状（如油污）的面积。因此，他们需要使用**估算**方法。

- **具体分析：** 题目要求“估计”油污的面积。在没有网格纸的情况下，最常见的估算方法是将不规则形状分解为更规则的近似形状（例如矩形和三角形）或使用叠加网格计数的方法。学生需要根据地图的比例尺（1 厘米代表 10 公里）进行目视判断和近似测量，意识到结果将是一个估计值，而不是精确值。这种对现实世界背景下所需准确度水平的理解指导了解决策略的选择。

2. 操作数字、图形和统计数据及信息、代数表达式和方程式以及几何表示 (列表中的第 7 点)

该任务要求学生操作多种类型的信息。

- **具体分析：** 学生需要从图形表示（地图）中提取信息（目视判断形状的尺寸），应用比例尺进行数字操作（将厘米转换为公里，然后计算平方公里），并将结果表示为最终的面积数值。这涉及在不同表示形式之间切换和操作数据。

3. 利用多步骤程序得出数学解、结论或概括 (列表中的第 12 点)

估算过程涉及多个步骤。

- **具体分析：**
  1. 选择一种估算方法（例如，将形状分解为几个近似的规则形状）。
  2. 目视测量或估计这些近似形状在图上的尺寸（厘米）。
  3. 应用比例尺将厘米尺寸转换为公里尺寸。
  4. 使用面积公式计算每个近似形状的实际面积（平方公里）。
  5. 将所有近似面积相加得出总的估计面积。
     这是一个清晰的多步骤过程。

4. 理解并利用基于定义、规则和形式系统的构建，包括运用熟悉的算法 (列表中的第 5 点)

学生需要理解面积和比例尺的基本数学定义和规则。

- **具体分析：** 学生必须理解比例尺如何影响面积单位的转换（1 cm^2 代表 10 km × 10 km = 100 km^2）。他们需要知道如何计算矩形或三角形的面积公式，并将这些基本算法应用于估算过程中。